

AJEET KUMAR GUPTA

Data Engineer | Python | Backend & Pipelines

Pune, India • +91 7440885722 • ajeetmalhar5@gmail.com • [GitHub](#) • [LinkedIn](#)

PROFESSIONAL SUMMARY

Data Engineer and Backend Developer with hands-on experience building scalable data pipelines, ETL systems, and production APIs in Python. Currently designing high-throughput ingestion and distributed processing systems at TVH using FastAPI, PostgreSQL, RabbitMQ, and GCP. Passionate about clean data modeling, reliable pipelines, and LLM-powered data tools like RAG and text-to-SQL that bring natural-language interfaces to structured and unstructured data.

TECHNICAL SKILLS

Languages/Backend: Python, SQL, FastAPI, REST APIs, Async, Multithreading

Data Engineering: ETL, Airflow, PySpark, Data Modeling, Batch & Stream Processing, RabbitMQ

Databases/Cloud: PostgreSQL, BigQuery, MySQL, MongoDB, Redis, SQLite, GCP, Docker

Data & GenAI: Pandas, NumPy, Scikit-learn, LangChain, RAG, OpenAI/Claude APIs

PROFESSIONAL EXPERIENCE

Data Engineer

Dec 2024 – Present

TVH India

- Built scalable ETL pipelines using **Python + FastAPI + Airflow** to ingest data into PostgreSQL & BigQuery
- Designed distributed processing system using **RabbitMQ (producer/consumer)** for async job execution
- Processed large datasets with **PySpark** and parallelized workflows using **async + multithreading**
- Optimized PostgreSQL queries (indexing, partitioning), improving pipeline performance
- Containerized services with Docker and deployed on **GCP**, ensuring reliability and monitoring

Python Developer

Mar 2023 – Dec 2024

PVKL Tech Services Pvt. Ltd. (Provakil)

- Integrated **15+ data sources** and built ETL pipelines for data cleaning, normalization & validation
- Reduced pipeline turnaround time from **7 days → 3 days** via parallelization & optimization
- Built monitoring system over **8,000+ sources**, reducing customer issues by **80%**

PROJECTS

Zomato NL→SQL Engine

Tech: Python, Fast API, SQLite, Claude API

- Built a natural-language-to-SQL analytics tool that lets non-technical users query a Zomato-style orders database (5,000 synthetic records) in plain English instead of writing SQL by hand.
- Designed a FastAPI backend that feeds DB schema as context to the LLM, executes generated SQL against SQLite, and returns structured JSON results; covered aggregations, time-based, and multi-condition queries with integration tests.

RAG PDF Question-Answering System

Tech: Lang Chain, Chroma DB, Sentence Transformers, Streamlit

- Built an end-to-end RAG pipeline that ingests PDFs, chunks with RecursiveCharacterTextSplitter, embeds locally with Sentence Transformers, and stores vectors in a persisted Chroma DB instance.
- Used MMR retrieval to improve answer diversity, implemented MD5-based file deduplication, and made LLM/embedding providers swappable (Anthropic/OpenAI, local/OpenAI) via .env config.

EDUCATION & ACHIEVEMENTS

B.Tech in Electrical Engineering, SGSITS, Indore — CGPA: 8.05 Jul 2018 – Jul 2022

Class 12 (HSC), Govt. Multipurpose Malharashram H.S. School, Indore — 90.00% Jun 2017 – May 2018

ACHIEVEMENTS & CERTIFICATION

Certifications: Data Scientist Certification — EdYoda Digital University (2022); Python Programming: A Concise Introduction.

Achievements: Cleared JEE Mains & Advanced (2018); District Merit Rank 1 (95.17%) in Class 10 — M.P. Government SUPER-100 program.